

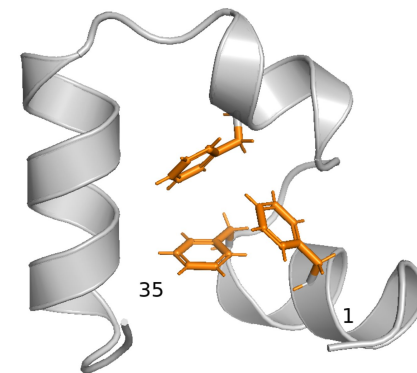
BIOLOGICAL DATASETS FOR COMPUTATIONAL PHYSICS

Markov State Modelling of the HP35 domain

Student: **Miriam Zara** - ID 2163328

Instructor: **Prof. Monika Fuxreiter, PhD**

Academic year 2025-2026

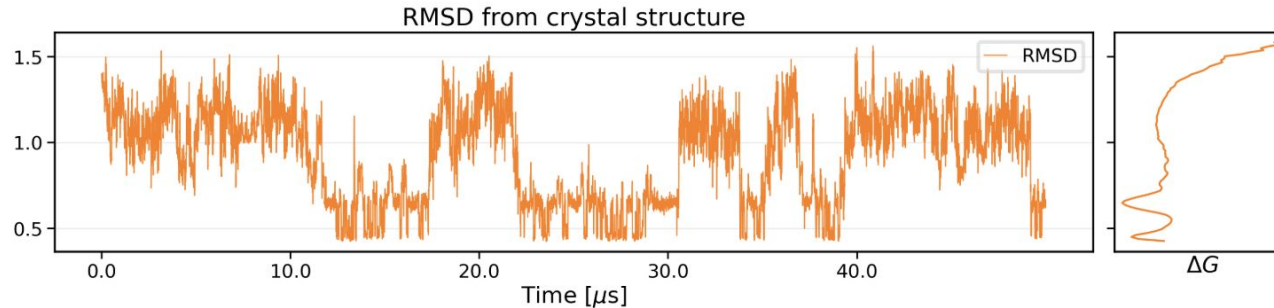


PDB 2F4K

J. Kubelka, T.K. Chiu, D.R. Davies et al.
Sub-microsecond protein folding, J. Mol. Biol. 2006

HP35 (Nle/Nle mutant)

- Villin headpiece subdomain (HP35) Nle/Nle mutant (35 residues)
- Ultra fast autonomously folding domain: < **1 μ s** (experimental), **$\approx$ 2-3 μ s** (simulation)
- Single **305 μ s** MD trajectory at T=**360 K** (near melting temperature $\rightarrow$ many folding/unfolding events)



Biological question: Does HP35 have resolvable **metastable substructure** (e.g. folding intermediates, misfolded states, alternative folded conformations)?

Markov State Modelling (MSM):

describe the biomolecular dynamics as a Markov Chain

Spectral analysis (PCCA++ algorithm):

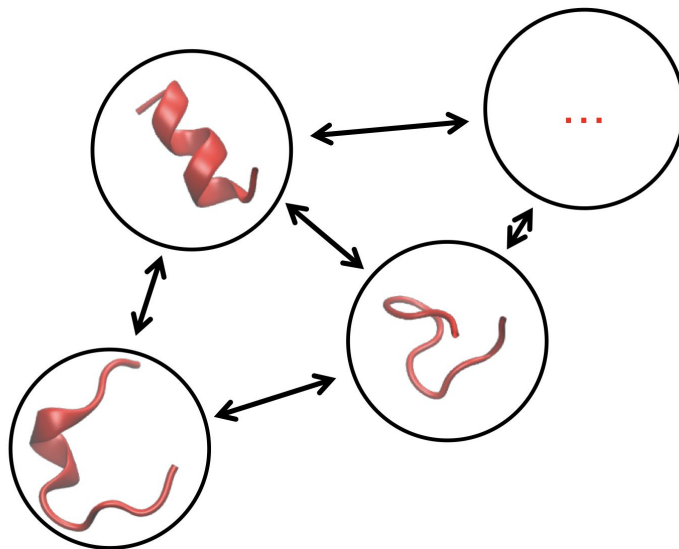
Transition matrix eigenvalue/eigenvector pairs:

- Eigenvector $\leftrightarrow$ probability exchange mode
- Eigenvalue $\leftrightarrow$ associated timescale

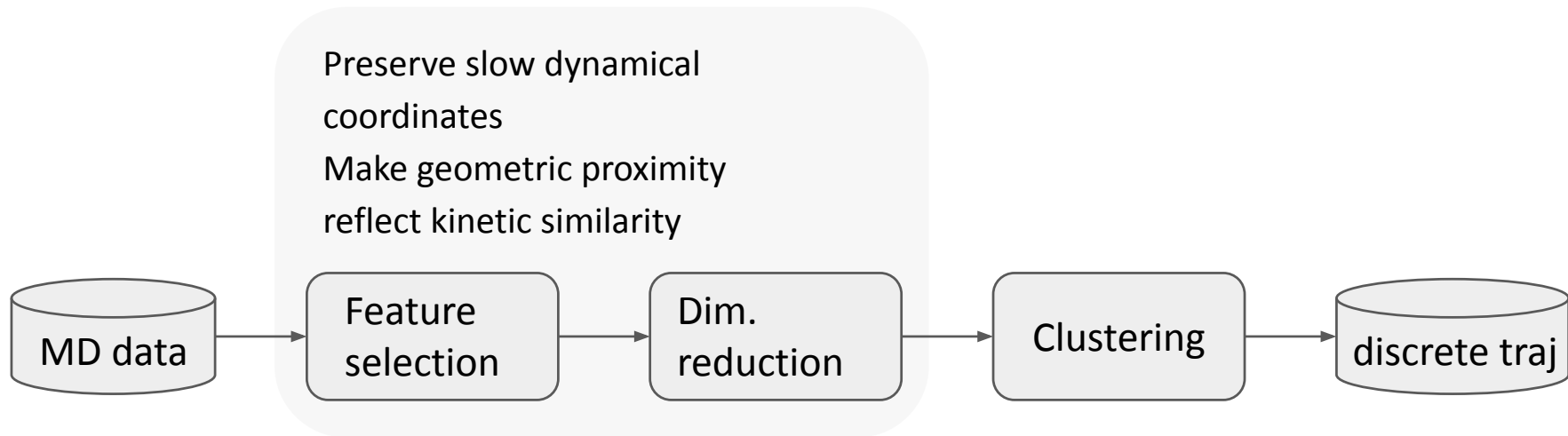
Coarse graining:

Microstates $\rightarrow$ macrostates $\equiv$ metastable states

$$T_{ij} = \mathbb{P}(s(t + \tau) = j \mid s(t) = i)$$



MSM construction: many reasonably valid choices at each stage



Suboptimal choices propagate error through the pipeline

Upstream decisions determine MSM quality and influence downstream analysis

Methodological question: How does the MSM pipeline affect the resulting macrostates?

Four MSM pipelines

Built combining **2** feature sets and **2** dim. reduction algorithms

PCA (maximize *variance*)
large motion = important motion

tICA (maximize *autocorrelation*)
slow motion = important motion

42 contact **distances**

pca-distances

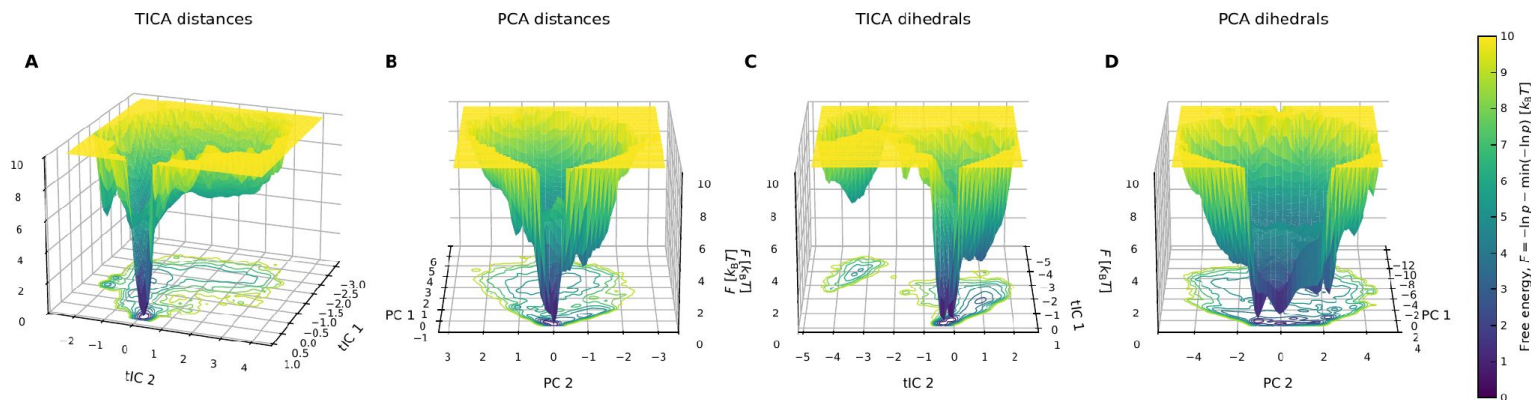
tica-distances

62 backbone **dihedrals**

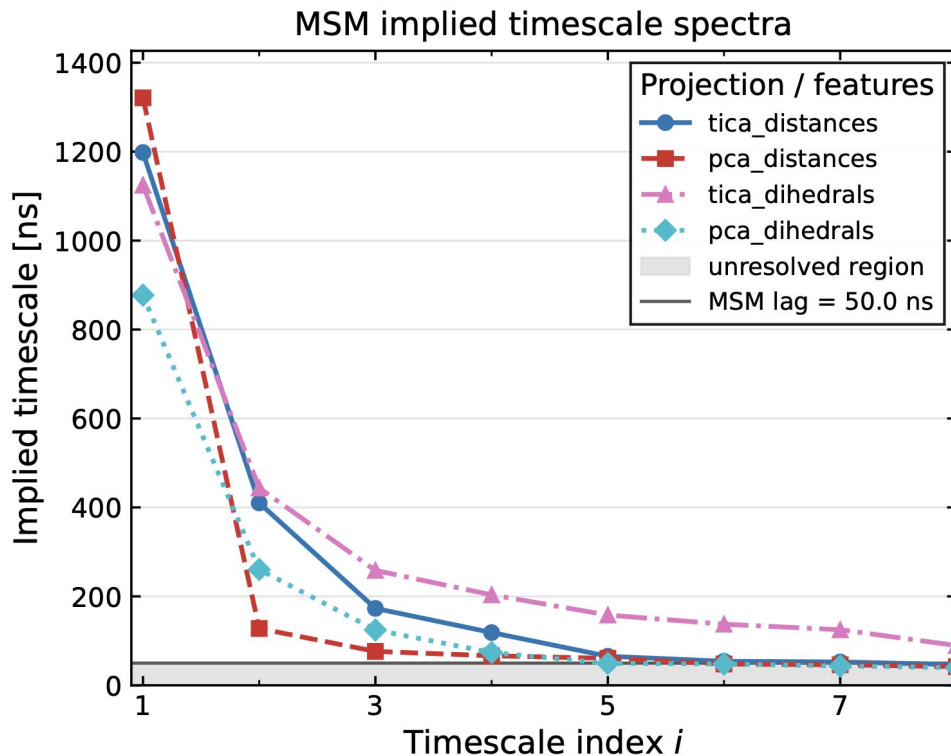
pca-dihedrals

tica-dihedrals

Free energy surface obtained by projection of trajectory on the first two reduced coordinates



The models already differ in their timescale spectra



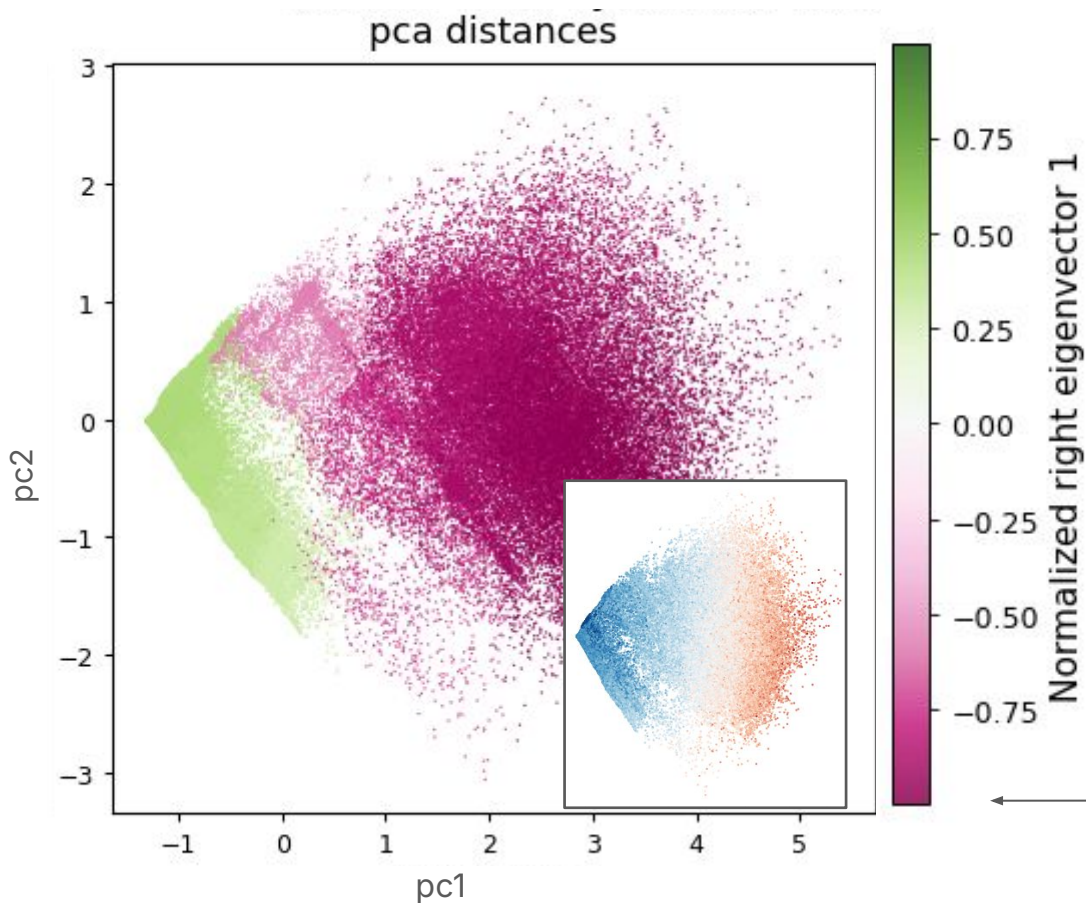
Each time scale corresponds to one slow dynamical process.

Large gaps indicate well-separated kinetic processes.

PCA-distances resolves essentially one dominant slow process

Other models resolve additional slow processes

The first eigenvector identifies the folding-unfolding transition



2 state PCCA++

All models recover nearly* the same two-state partition

31–32%

unfolded-side

67–68%

folded-side

consistent with Piana et al. for this trajectory

* % frames placed on opposite sides of the boundary: ≈ 2
% tica-distances/pca-distances
 ≈ 7 -8% tica-dihedrals vs others

Large: Sign pattern of first right eigenvector
Probability flows between states with opposite sign
Small: Pattern of RMSD from crystal structure

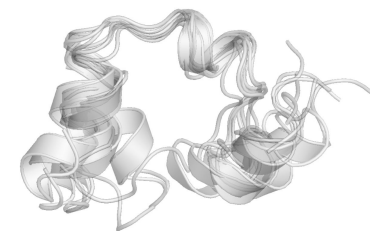
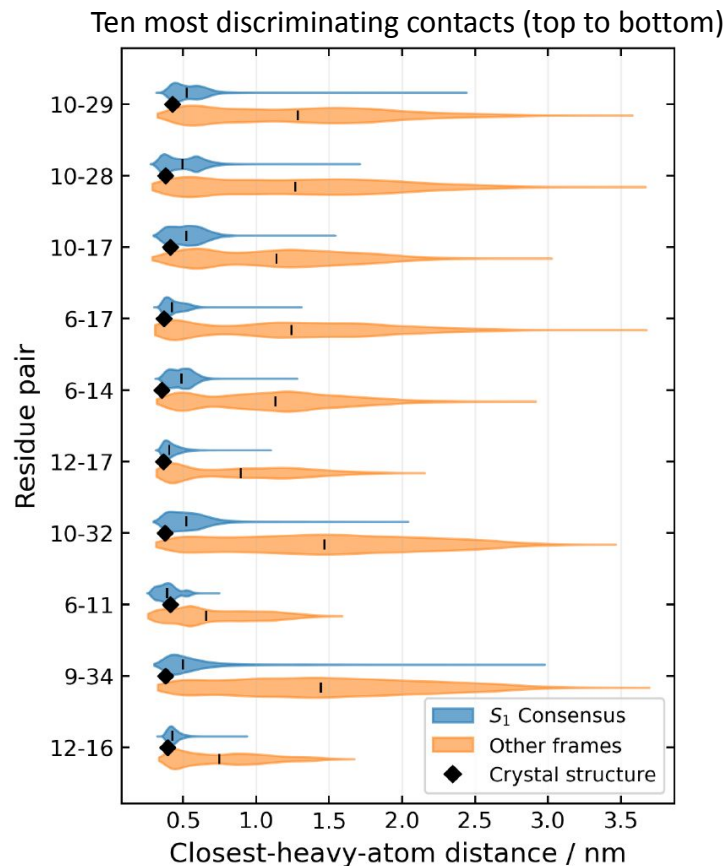
Folded-side basin is distinguished by hydrophobic core and close packing

Method: ranked every native contact by how sharply it separates the two macrostates

Result:

- Key contacts involve Phe 6, Phe 10, Phe 17 (hydrophobic core)
- Intra-helix contacts do not appear -> helix formation frequent in the unfolded macrostate

$$\text{score}_{i,j} = \frac{\text{median}(r_{ij}^{\text{other}}) - \text{median}(r_{ij}^{\text{consensus}})}{[IQR_{ij}^{\text{other}} + IQR_{ij}^{\text{consensus}}] / 2}$$



Overlay of 8 random frames from the S₁ macrostate

Only the tICA-distances model identifies a compact non-native state

3 state PCCA++ on tica-distances splits the unfolded basin
revealing well characterized new state

Population $\approx 1\%$

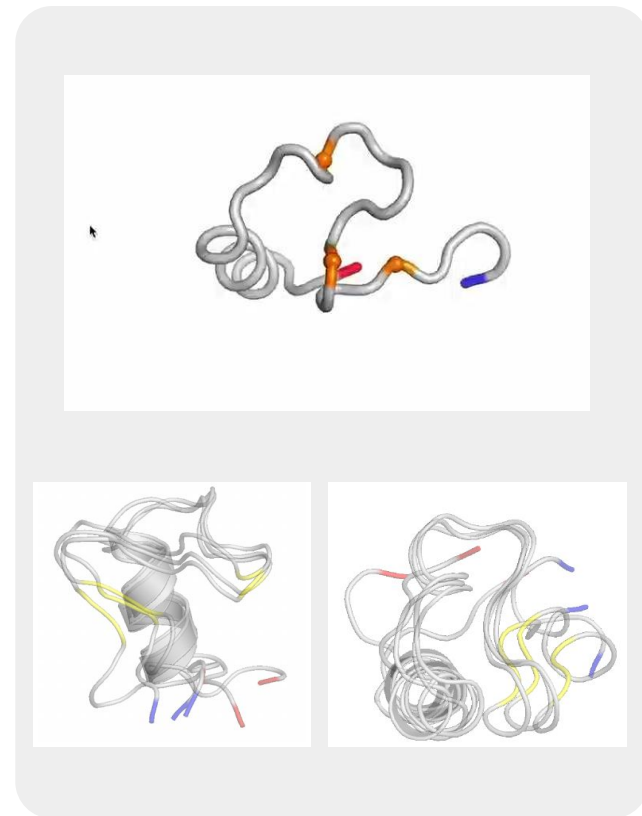
Lifetime $\approx 1 \mu\text{s}$

Rare but
long-lived

Structural signature

- Helix 3 stays folded
- Stable conformation of residues 15-21 (second loop)
- residues 6–14 form a stable hairpin-like segment in place of helix 1.

Pipeline dependence: not identified by the three other models

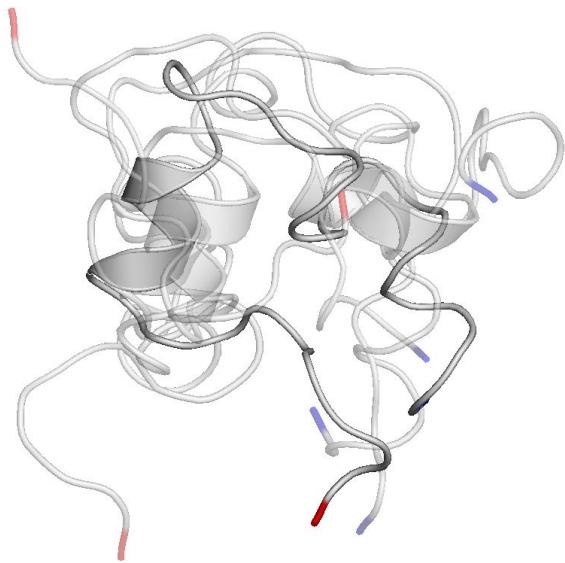


Unlike tica-distances, tica-dihedrals resolves a local loop rearrangement

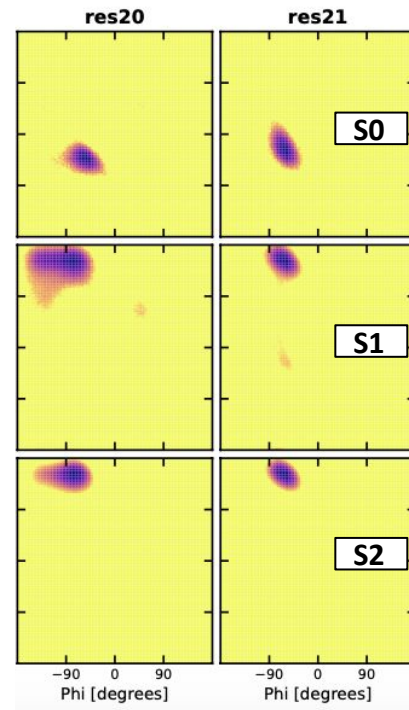
3-state PCCA++ on
tICA-dihedrals splits the
unfolded basin revealing a
state with poor structural
characterization

Population $\approx 6\%$
Lifetime $\approx 0.32 \mu\text{s}$ **S0**

The associated slow process
describes a local loop
rearrangement within the
unfolded ensemble



*Overlay of 4 random frames
from the S0 macrostate*



*Residues 20 and 21 switch between
alternative backbone conformations*

Main Findings

- All models recover folding-unfolding
- **tICA-distances** identifies a metastable misfolded state
- Different pipelines resolve different slow processes

Avenues to explore for future work

- More systematic comparison of PCCA++ partitionings
- Kinetic analysis: pathways and fluxes (Transition Path Theory)

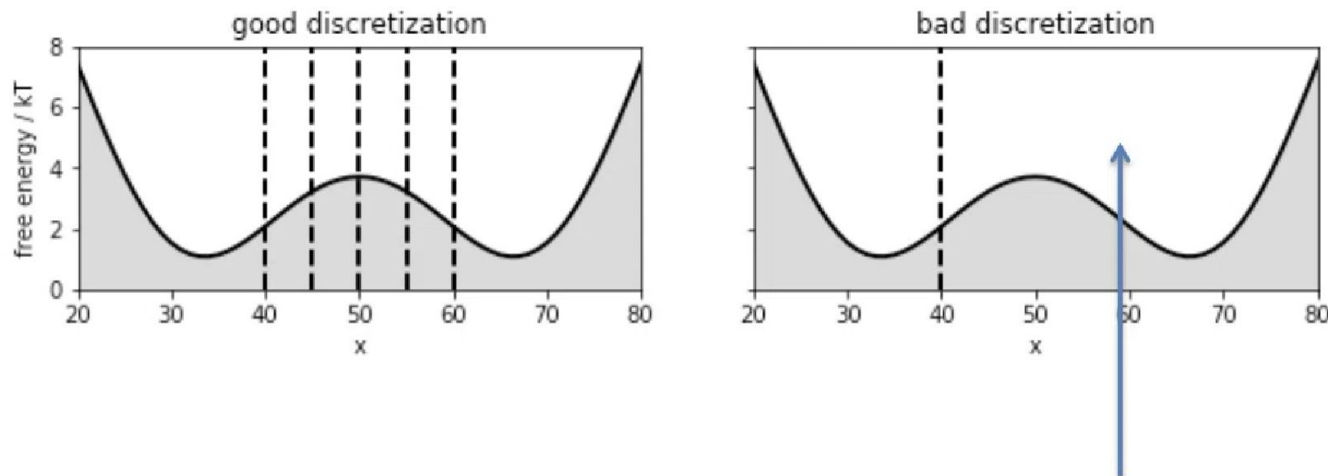
Thank you for the attention

Data github.com/moldyn/HP35-DESRES

Code github.com/miriamzara/MSM

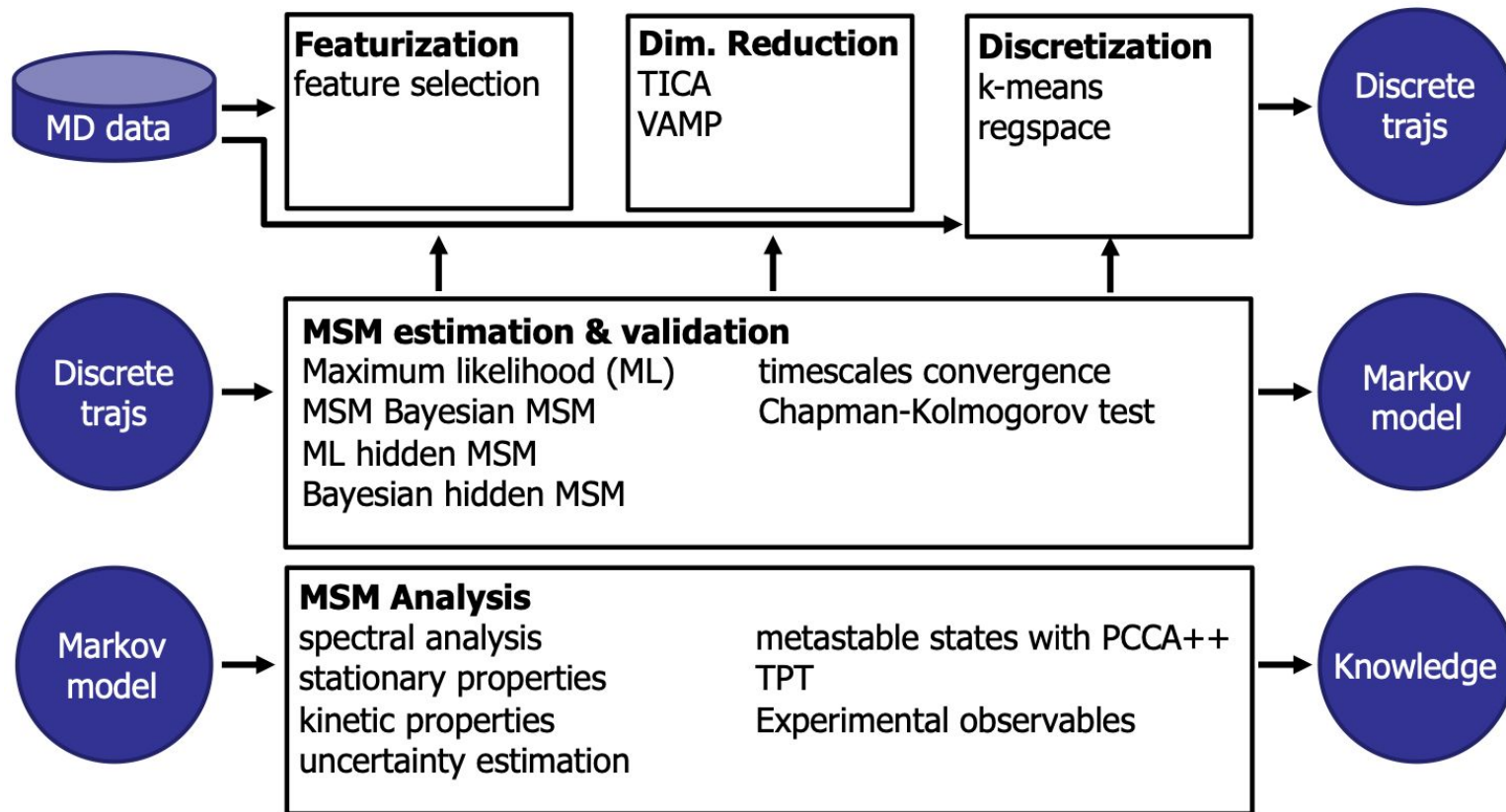
Backup slides

Bad discretization breaks the Markovianity assumption



In the “bad discretization” case, the barrier is embedded in one of the states. This generates a “long term memory” effect: the rightmost state could actually be short-lived (if we are on the left of the barrier) or long-lived (if we are on its right). These two cases are convoluted into the same, so that the present state information itself is not sufficient to predict the “future” of the system any more.

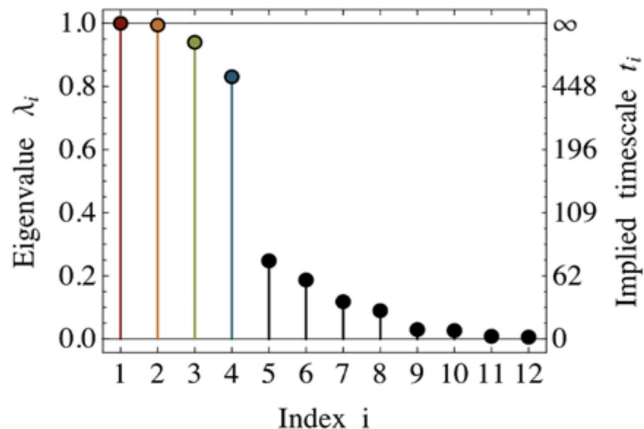
Markov Modelling Workflow



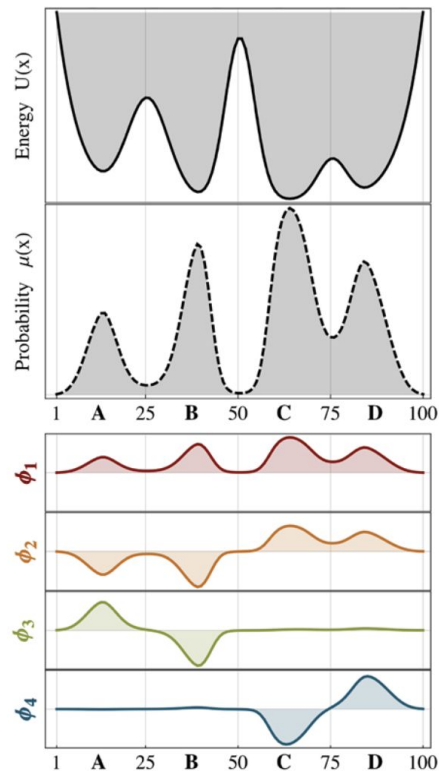
From FU Berlin,
PyEmma
Workshop

<https://ftp.mi.fu-berlin.de/pub/ag-cmb/pyemma-workshop-2019/slides/intro.pdf>

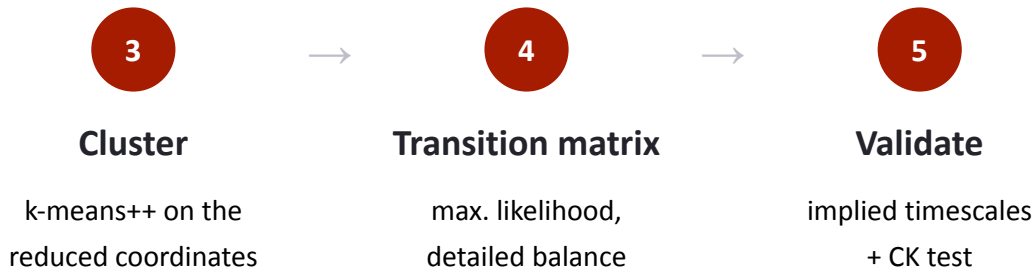
PCCA++ identifies metastable states using the sign pattern of dominant eigenvectors



From: G.R. Bowman, V.S. Pande and F. Noe,
*An Introduction to Markov State Models and
 Their Application to Long Timescale
 Molecular Simulation* Springer (2013)



Common parameters across pipelines: Markov lag and Cluster number



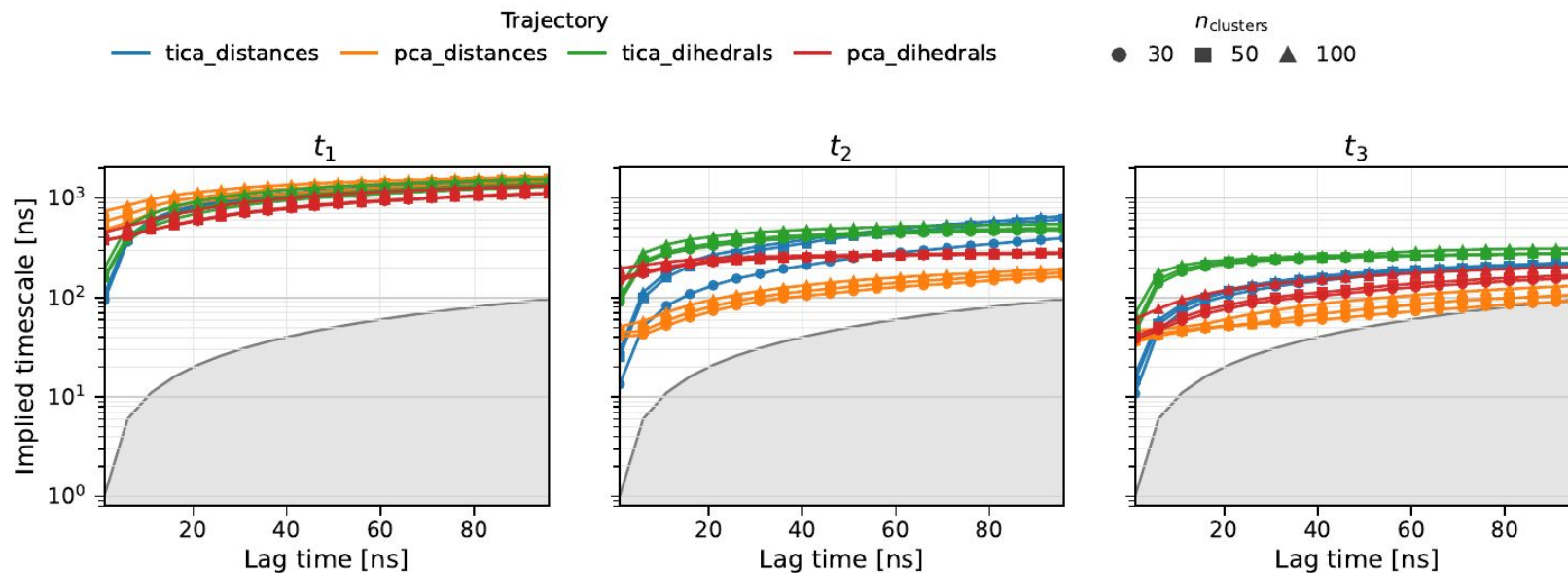
Some points:

- More clusters give resolution but fewer transitions per state and a network that can disconnect.
- The lag must be long enough for Markovianity yet well below the folding timescale (μ s).
- Common parameters: all four pipelines share the same k and τ , so whatever differs between the models comes from the features and the reduction — not from the discretisation.

Parameters used across pipelines:

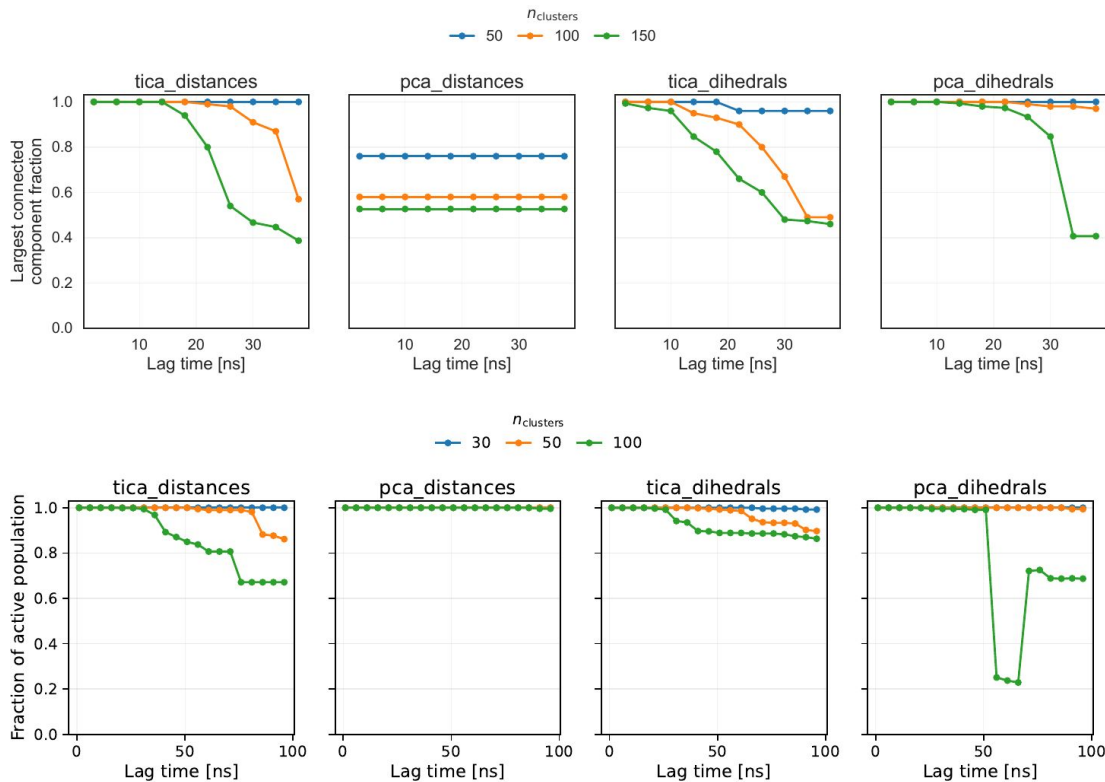
1. **Number of clusters: $k=50$**
scanned over $\{30, 40, 50\}$
2. **Markov lag: $\tau=100$**
scanned over $[10, 1000]$

Choice of number of clusters and lag time



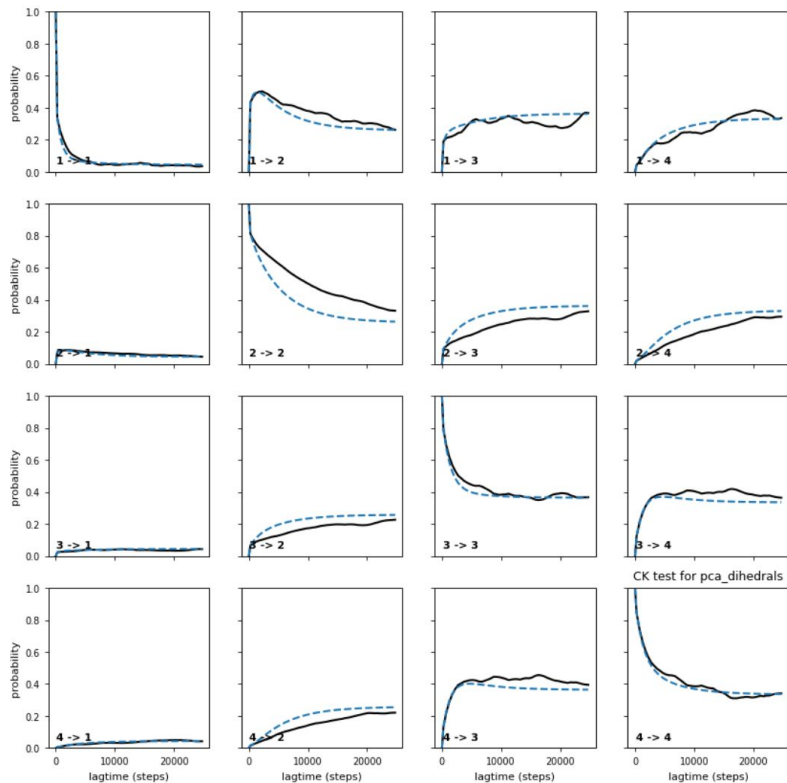
Top: implied timescales vs lag (SF2). Bottom left: largest connected component and active population vs lag (SF3).

Choice of number of clusters and lag time



Top: implied timescales vs lag (SF2). Bottom left: largest connected component and active population vs lag (SF3).

Chapman–Kolmogorov test

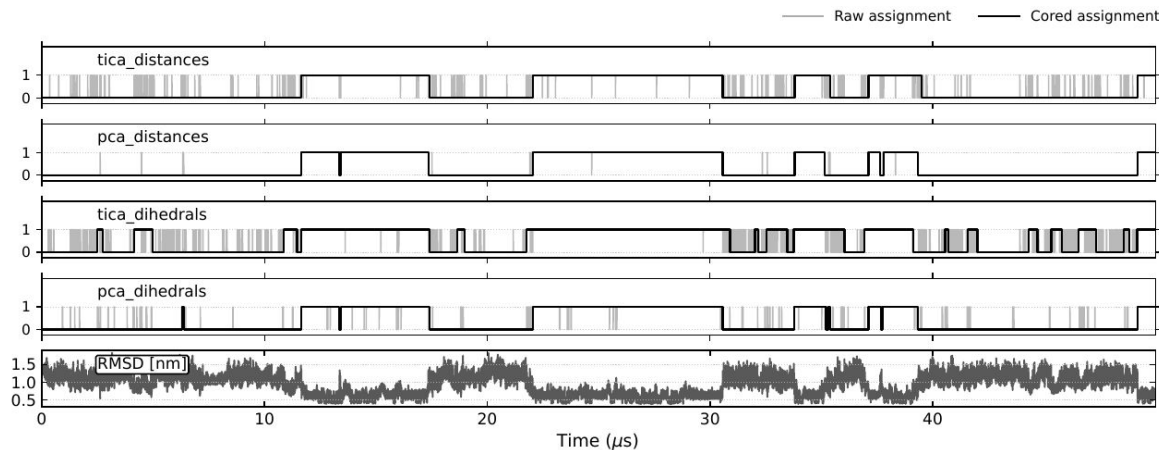


Shown for pca-dihedrals; the other three behave similarly.

The test:

- The microstate MSM at $\tau = 50$ ns is lumped into four metastable sets.
- For a range of lag times $k\tau$, the transition probabilities **predicted** by the $\tau = 50$ ns model (dashed) are compared with those **estimated independently** at each lag (solid).
- The two track closely and relax to consistent stationary values, so the model is approximately Markovian at the chosen lag.

Dynamical coring removes boundary recrossings

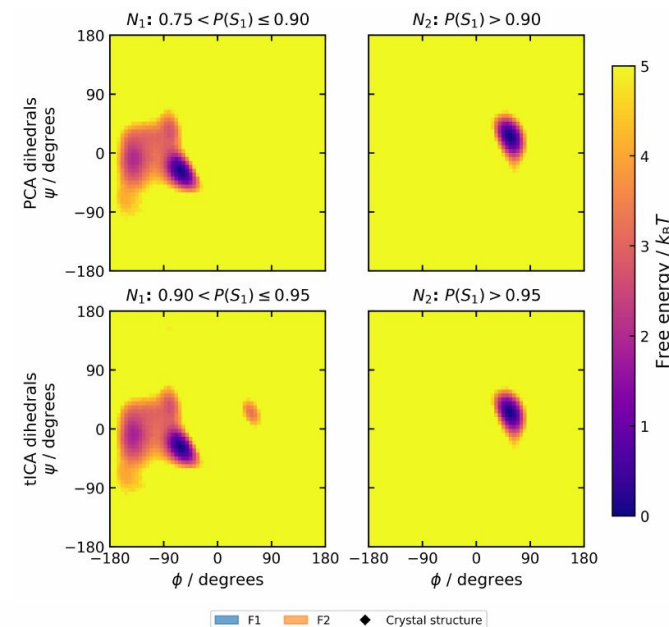
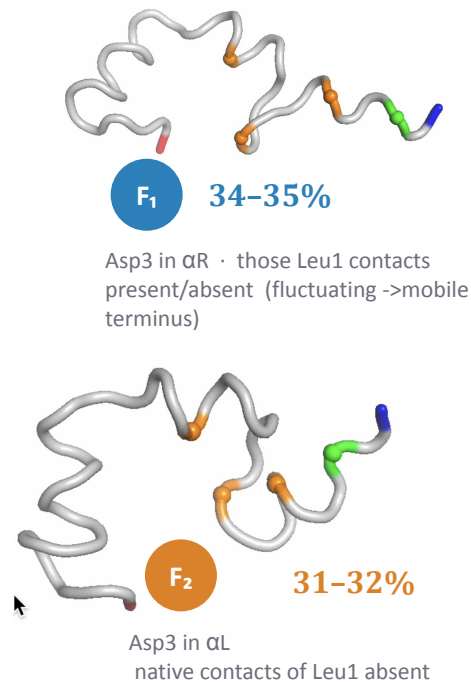
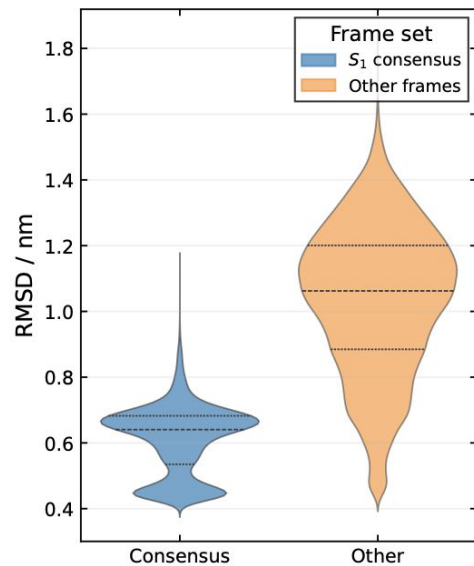


The problem. Assigning every frame to its highest-membership macrostate produces rapid recrossings near a state boundary. These inflate the apparent number of transitions, and mostly reflect boundary noise rather than genuine interstate crossings.

The fix. A transition into a new macrostate is registered only once the trajectory stays there for at least $t_{\min} = \tau = 50$ ns, so only long-lived occupancies survive. Grey = raw assignment, black = cored. All lifetimes and empirical populations quoted here are computed from the cored trajectory.

Inside the folded-side basin

RMSD, the 10–29 contact and the membership distributions of both dihedral models are all bimodal inside the folded basin — yet PCCA++ with three states does not split it. So I subdivided it by hand

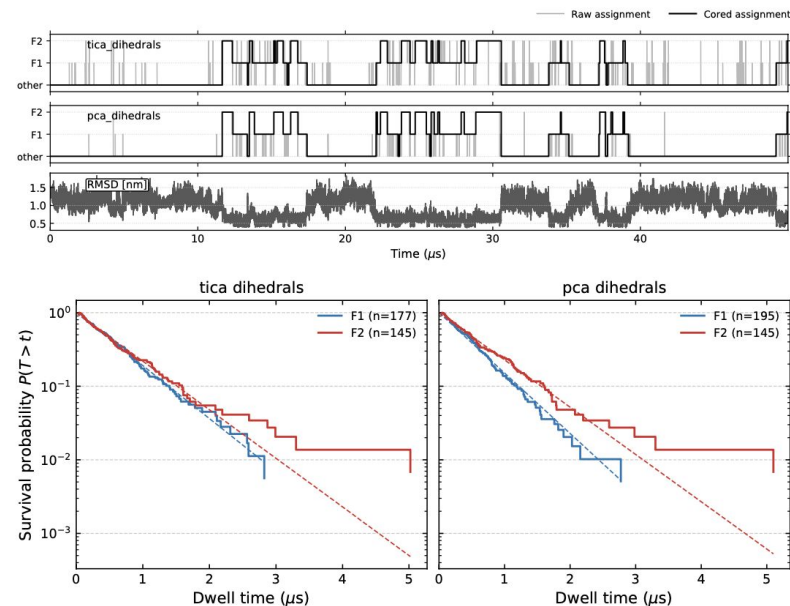


(ϕ, ψ) of Asp3 in the two subpopulations, for the PCA (top) and tICA (bottom) dihedral models. The whole distinction rests on one residue.

F_1 and F_2 lifetimes

Model	State	N visits	Mean [μ s]	Median [μ s]	Population
tica-dihedrals	F_1	177	0.603	0.463	0.350
tica-dihedrals	F_2	145	0.658	0.435	0.313
pca-dihedrals	F_1	195	0.529	0.412	0.338
pca-dihedrals	F_2	145	0.676	0.480	0.321

Residence times in the two folded sub-basins, from the cored microstate trajectory of each dihedral model.



Survival probability $P(T > t)$ with an exponential fit, for both dihedral models.